Internet Research

Balancing efficiency and responsibility in generative AI adoption

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Abstract

Purpose – This study investigates how firms achieve sustainable outcomes from generative artificial intelligence (GenAI) adoption by balancing efficiency gains with ethical and environmental responsibilities. It integrates dynamic capability theory and responsible AI governance to examine the mediating role of digital dynamic capability (DsDC) and the moderating effect of responsible AI orientation (RAIO).

Design/methodology/approach – Two complementary empirical studies were undertaken. Study 1 applies a staggered difference-in-differences design with propensity-score matching (DiD-PSM) to 500 GenAI adoption events (2019–2024). Study 2 employs a two-step system generalised method of moments (GMM) estimator for 1,274 firm-year observations (2016–2024) to analyse persistence, mediation and moderation effects.

Findings – GenAI adoption was associated with a market-perceived efficiency signal (a 2.7% rise in cumulative abnormal returns) alongside a 1.9% decline in environmental, social and governance sentiment, a reputational rather than substantive sustainability indicator, evidencing an efficiency-responsibility tension. Over time, DDC helped partially translate these short-term gains into sustainable performance, while RAIO mitigated ethical and reputational risks, enhancing firms' long-term legitimacy.

Research limitations/implications – The study relies on archival indicators and text-derived governance measures; future research could combine these data with surveys or internal audit evidence.

Practical implications – Managers should pair GenAI deployment with adaptive digital capability and responsible AI governance to protect stakeholder trust.

Originality/value – This study advances theory by integrating capability and governance perspectives to explain the sustainability of GenAI adoption. Methodologically, it introduces temporal triangulation by combining DiD-PSM and System GMM, offering a novel dynamic causal framework for responsible digital transformation.

Keywords Generative AI, Digital dynamic capability, Responsible AI orientation, Sustainability

Paper type Research article

1. Introduction

Generative artificial intelligence (GenAI) has evolved from experimental models into a general-purpose technology reshaping business processes, creativity and decision-making. Leveraging transformer architectures, GenAI synthesises text, images, code and multimodal outputs at scale, transforming how firms sense and respond to their environments. Embedded in customer interaction, knowledge management and design workflows, GenAI turns generativity into an organisational resource that drives productivity and recombinative innovation (Dwivedi *et al.*, 2023; Ooi *et al.*, 2025). By lowering content and ideation costs, GenAI accelerates experimentation and decision cycles, enables faster design and problem-solving and strengthens firms' capacity to convert unstructured data into actionable outputs. Yet its organisational implications are not uniformly beneficial. While GenAI improves efficiency, it also introduces vulnerabilities: energy-intensive computation raises environmental concerns, algorithmic opacity impedes accountability and synthetic media threatens informational integrity and trust (Bengio *et al.*, 2024; Campbell *et al.*, 2022). Accordingly, market-perceived efficiency signals enabled by AI (efficiency gains, hereafter,



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as perceived by investors and stakeholders rather than directly observed operational outcomes) may coexist with environmental, social and governance (ESG) externalities, creating a structural tension rather than a linear performance effect. This duality gives rise to what we conceptualise as the GenAI sustainability paradox: a structural organisational tension in which the same generative mechanisms that drive short-term efficiency and innovation simultaneously generate longer-term ethical, social, environmental and legitimacy risks.

If sustainability entails balancing profit, planet and people, the central challenge is not simply whether GenAI improves performance, but how firms convert short-term efficiency into durable and responsible performance. GenAI intensifies this challenge because firms are adopting powerful generative tools under conditions of rapid diffusion, regulatory uncertainty and still-evolving governance arrangements. Prior research has largely examined either the performance benefits of AI or its ethical risks in isolation, offering limited insight into how organisations reconcile these competing forces over time (Dwivedi *et al.*, 2023; Susarla *et al.*, 2023). What remains under-theorised is the organisational mechanism through which efficiency gains can be aligned with sustainability and legitimacy. This study proposes two complementary levers. The first is digital dynamic capability (DDC), defined as the ability to reconfigure digital assets and learning routines for adaptive transformation (Mikalef and Pateli, 2017). The second is responsible AI orientation (RAIO), which reflects the institutionalisation of transparency, fairness, accountability and human oversight in AI-related organisational practices (Abboud *et al.*, 2026; Floridi and Cowls, 2022). We argue that navigating the GenAI sustainability paradox requires both adaptive capability and governance orientation: DDC enables firms to transform digital resources and learning routines, whereas RAIO helps ensure that such transformation remains ethically bounded and socially legitimate. This perspective therefore conceptualises sustainability not as a direct outcome of GenAI adoption, but as a mediated and governed process of organisational transformation.

Despite growing commentary on GenAI, empirical evidence on how firms sustain GenAI-driven value remains scarce. Few studies trace whether ESG and reputational outcomes persist, deteriorate, or recover after adoption. Existing work tends to isolate either short-term market reactions or cross-sectional correlations, lacking designs that combine credible causal identification with dynamic modelling over time (Susarla *et al.*, 2023). To address this gap, this study employs a two-stage design. First, a staggered difference-in-differences with propensity-score matching (difference-in-differences design with propensity-score matching [DiD-PSM]) is used to estimate the short-run effects of GenAI adoption on market and sustainability signals. Second, a two-step system generalised method of moments (GMM) analysis is used to trace longer-run sustainability dynamics, including persistence, mediation and moderation effects. As these two designs necessarily draw on different samples, time horizons and outcome measures, they are deliberately structured as complementary components of a single temporal-triangulation strategy rather than a single study split across two samples; the rationale for this design choice is elaborated at the end of Section 4. Guided by this design, the study addresses three research questions.

RQ1. How does GenAI adoption affect firms' short-term efficiency and sustainability performance?

RQ2. To what extent does DDC mediate the relationship between GenAI adoption and long-term sustainability performance?

RQ3. Does responsible AI orientation (RAIO) moderate the efficiency–sustainability relationship, strengthening the positive and mitigating the negative effects of GenAI adoption?

Integrating dynamic capability theory with responsible AI governance, we argue that the effects of GenAI on sustainability are not technology-determined but contingent upon organisational capabilities and governance structures.

This study makes three contributions. First, it links capability and governance perspectives to explain how firms may mitigate, rather than eliminate, the GenAI sustainability paradox.

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Second, it combines DiD-PSM and System GMM to capture immediate effects and longer-run adaptation. Third, it identifies the organisational conditions under which GenAI can support sustainable outcomes. These contributions address the distinctive governance concerns of generative models, including synthetic content, hallucination and transparency. Building on Internet Research scholarship on digital transformation and AI governance (Birkstedt *et al.*, 2023; Dwivedi *et al.*, 2023), the study provides causal, longitudinal evidence on the efficiency-responsibility tension. It extends this literature by jointly modelling DDC and RAIO, combining short- and long-term analyses and identifying when RAIO converts efficiency into durable, legitimate performance.

2. Theoretical background

This study integrates four literature streams, each serving a distinct theoretical function. First, research on GenAI's bright and dark sides defines the focal paradox: the same technology can generate productivity gains and ethical, environmental and legitimacy risks (Chin *et al.*, 2025; Dwivedi *et al.*, 2023). Second, sustainability and ESG scholarship establishes the outcome domain, showing that AI-enabled efficiency does not automatically produce durable ESG outcomes (George *et al.*, 2021; Wu *et al.*, 2026). Third, dynamic capability theory explains how firms convert technological potential into adaptive outcomes; DDC operationalises the sensing-seizing-transforming logic in the GenAI context (Mikalef and Pateli, 2017; Teece, 2018). Fourth, responsible AI governance specifies when such reconfiguration remains ethically bounded and socially legitimate; RAIO captures this condition at the firm level (Balasubramaniam *et al.*, 2023; Floridi and Cowls, 2022). Together, these streams position the GenAI sustainability paradox as the focal phenomenon, sustainability as the outcome domain, DDC as the organisational mechanism and RAIO as the governance condition – an architecture not previously assembled within a single causal model.

The Online Appendix provides a detailed positioning of this framework relative to technology-adoption research and summarises the core literature streams (Section A1 and Table OA1). This study advances prior work in three ways. First, whereas GenAI and sustainability studies often treat capability and governance separately (Bag *et al.*, 2026; Wu *et al.*, 2026), this study jointly tests DDC as a mediator and RAIO as a moderator. Second, it introduces a normative governance construct into dynamic capability research. Third, its temporal triangulation of DiD-PSM and system GMM captures immediate effects and subsequent adaptation that cross-sectional designs cannot observe (Fosso Wamba *et al.*, 2024; Xia *et al.*, 2026).

2.1 Generative AI and the sustainability paradox

GenAI represents a transformative class of machine learning systems capable of autonomously producing novel and contextually adaptive outputs, including text, images, code and multimodal artefacts from vast datasets. Unlike traditional analytic AI, which classifies or predicts within fixed boundaries, GenAI recombines learned representations to create new possibilities. This adaptive capacity marks GenAI as a general-purpose technology reshaping digital business architectures, creativity and decision-making (Dwivedi *et al.*, 2023; Ooi *et al.*, 2025).

GenAI's diffusion presents a paradox: it accelerates efficiency and innovation while simultaneously threatening sustainability. On its bright side, GenAI enhances scalability, productivity and creativity. It reduces production costs and cycle times, facilitates "augmented ideation" (Eapen *et al.*, 2023) and allows smaller firms to compete in complex design and innovation tasks once dominated by large incumbents (Ooi *et al.*, 2025). However, GenAI's dark side is equally profound. As Bender *et al.* (2021) warned, generative models, or "stochastic parrots", can replicate bias and misinformation under a veneer of coherence (Agarwal *et al.*, 2026; Chua *et al.*, 2025). Their opacity undermines accountability, while synthetic content erodes informational integrity and public trust (Abboud *et al.*, 2026). Bengio

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et al. (2024) further highlighted the environmental toll: the carbon footprint of training large models rivals that of industrial operations, raising doubts about GenAI's ecological viability. Recent paradox-oriented AI research similarly suggests that emerging AI phenomena often produce simultaneous benefits and harms rather than unidirectional effects, reinforcing the need to theorise contradiction as structural rather than incidental (Chin *et al.*, 2025).

The Online Appendix elaborates the structural character of the "technological externality paradox" (Section A2). This issue is especially salient in digital business, where rapid GenAI diffusion coincides with demands for algorithmic transparency, responsible data use and lower environmental impact. The GenAI sustainability paradox is therefore a digital-transformation governance challenge, not merely a technology-adoption problem. It also aligns with Internet Research's focus on misinformation, content authenticity and platform accountability (Abboud *et al.*, 2026). This study addresses that agenda by testing whether RAIO moderates the sustainability consequences of GenAI adoption in digitally intensive firms.

Terminological note: This study employs three related but distinct terms that should not be conflated. "GenAI sustainability paradox" is the primary theoretical construct, referring to the structural organisational tension in which the same generative mechanisms that drive short-term efficiency and innovation simultaneously generate longer-term ethical, social, environmental and legitimacy risks; it is used as the overarching label throughout this manuscript. "Efficiency-responsibility tension" (and, equivalently, "efficiency-sustainability tension") denotes the operational manifestation of this paradox as observed empirically, specifically the simultaneous positive cumulative abnormal returns and negative ESG sentiment documented in Study 1. "Technological externality paradox" (following Chin *et al.*, 2025) is used exclusively in this section to characterise the broader rebound-effect logic whereby process-level efficiency gains generate systemic externalities such as overproduction, cognitive dependency and environmental strain; it functions as a theoretical mechanism that explains why the GenAI sustainability paradox arises, not as a synonym for it.

2.2 Digital dynamic capability

In an era of rapid digital evolution, competitive advantage no longer depends on owning advanced technologies but on the capacity to sense, seize and transform opportunities, encapsulated by DDC (Mikalef and Pateli, 2017). DDC represents a higher-order capability that enables organisations to reconfigure digital resources and competencies in response to shifting technological and market conditions. Within GenAI, DDC determines whether adoption yields transient efficiency or enduring, responsible innovation. Rooted in Teece's (2018) dynamic capability theory, DDC reflects a firm's ability to integrate and reconfigure competences under change. Extending this logic, Mikalef *et al.* (2022) described DDC as the agility with which firms mobilise infrastructures, data assets and analytic expertise for adaptive transformation. As GenAI disrupts content creation, knowledge management and customer engagement, firms lacking DDC struggle to align novelty with strategic coherence and ethical accountability. This is especially important in AI contexts because value creation depends not simply on deployment but on the continuous reconfiguration of socio-technical systems under uncertainty (Kumar *et al.*, 2023; Vaillant *et al.*, 2025).

Mechanistically, DDC enables firms to embed AI tools into modular digital architectures, translate generative insights into cross-functional change and sustain experimentation and ethical learning over time. Empirical evidence shows that high-DDC firms achieve stronger returns on digital investment, faster innovation and greater resilience (Mikalef and Pateli, 2017; Mikalef *et al.*, 2022). Observable proxies include digital R&D intensity, flexible information technology (IT) investments, data reconfiguration agility and digital learning initiatives such as AI ethics or reskilling programmes. These reflect how DDC converts technological potential into adaptive, responsible performance.

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A discussion of DDC relative to adjacent organisational constructs, including absorptive capacity, organisational learning and governance/ethics maturity frameworks, is provided in the Online Appendix (Section A3). DDC is theoretically preferable here because GenAI creates not only a knowledge or compliance challenge but also an ongoing reconfiguration challenge under technological, social and regulatory uncertainty. A concluding discussion of the dynamic DDC–sustainability relationship is provided in the Online Appendix (Section A4).

2.3 Responsible AI orientation

RAIO builds upon the ethical principles of fairness, accountability, transparency and human oversight articulated by Floridi and Cowls (2022). Rather than a purely normative aspiration, RAIO reflects an organisational disposition to embed these principles in technological and managerial practice. RAIO embodies the capacity of a firm to operationalise ethical intent through institutional routines that govern data use, algorithmic design and human-machine collaboration. As a strategic orientation, RAIO manifests in governance frameworks that align ethical safeguards with innovation and risk control mechanisms. Rather than functioning as an external compliance checklist, it permeates the decision-making architecture of firms – informing how generative models are trained, audited and deployed to mitigate societal risks and foster digital responsibility (Abboud *et al.*, 2026; Agarwal *et al.*, 2026). In high-RAIO firms, responsible AI is internalised through leadership accountability, cross-functional governance and continuous review mechanisms that align technological development with stakeholder expectations.

RAIO should also be distinguished from adjacent notions such as AI governance, ethics maturity or explainability maturity. AI governance typically refers to the formal structures, policies and control mechanisms through which AI systems are monitored and regulated, whereas ethics or explainability maturity models evaluate the extent to which such practices are formalised and advanced across stages of organisational development (Balasubramaniam *et al.*, 2023). RAIO, by contrast, captures a firm-level strategic orientation: the extent to which responsible AI principles are internalised and enacted across managerial and technological decisions. Where governance maturity asks whether formal mechanisms exist and how developed they are, RAIO asks whether responsibility functions as an orienting logic that shapes AI-enabled transformation. This distinction is important because this study does not treat responsibility as a static compliance attribute but as a governance orientation that conditions the consequences of adaptive reconfiguration.

The moderating logic of RAIO in this study lies in its ability to recalibrate the balance between efficiency and sustainability. Firms characterised by strong RAIO are better positioned to mitigate the unintended consequences of GenAI, such as algorithmic bias, opacity, or excessive resource consumption, by embedding reflexive feedback loops and ethical evaluation checkpoints within their AI lifecycle. Consequently, RAIO enhances the transformative potential of DDC by ensuring that the processes of sensing, seizing and transforming digital opportunities are bounded by ethical responsibility. Taken together, DDC and RAIO explain complementary dimensions of firm-GenAI interaction: DDC captures adaptive reconfiguration, whereas RAIO explains how that reconfiguration is normatively bounded and rendered socially legitimate.

3. Hypotheses development

GenAI constitutes a class of technologies designed to automate creativity, augment cognitive work and accelerate decision-making across digital business functions. When integrated into organisational processes, GenAI systems streamline production, marketing, service and design by reducing cognitive load and labour redundancy. This automation generates measurable efficiency gains, including shorter development cycles, faster customer response

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times and improved throughput of informational tasks. Studies in digital innovation have shown that intelligent automation produces statistically significant improvements in process efficiency and cost-effectiveness, particularly when supported by pre-existing digital infrastructures (Dwivedi *et al.*, 2023; Ooi *et al.*, 2025). From the standpoint of dynamic capability theory, such technologies enhance the seizing dimension by enabling firms to capitalise on data-driven opportunities faster than competitors. At the same time, within the paradox framework of this study, these efficiency gains represent the initial and more visible organisational benefit of GenAI adoption rather than evidence that adoption automatically produces sustainable outcomes. Accordingly, it is reasonable to expect that firms adopting GenAI will experience an observable increase in efficiency performance relative to non-adopters. Therefore, this study proposes:

H1. GenAI adoption positively affects firm efficiency.

While efficiency has long been a central metric of technological progress, its relationship with sustainability remains contested. Classical economic reasoning suggests that resource efficiency contributes to sustainability by reducing waste and enabling cleaner production (Elkington, 2001). In digital business, efficiency improvements arising from AI automation can similarly reduce the carbon and material intensity of operations, thereby enhancing environmental sustainability. Moreover, efficient processes allow firms to reallocate saved resources towards social and governance initiatives, aligning economic performance with stakeholder expectations. Empirical studies have demonstrated that digital efficiency improvements often precede increases in ESG scores and stakeholder trust when accompanied by managerial commitment to sustainable practices (George *et al.*, 2021; Mikalef *et al.*, 2022). However, this positive association is conditional: efficiency supports sustainability only when firms reinvest those gains rather than exploit them for short-term optimisation alone. In other words, efficiency should be understood here as an enabling condition rather than a sufficient outcome in itself. Therefore, this study proposes:

H2. Efficiency gains positively influence sustainability performance.

Although GenAI adoption can yield immediate operational benefits, sustainable value creation requires firms to integrate, recombine and renew digital resources as environmental and ethical demands evolve. DDC provides this mechanism by enabling firms to sense technological opportunities, seize them through targeted investments and transform them into adaptive configurations (Mikalef and Pateli, 2017). In the GenAI context, sensing identifies efficiency gains and emerging ESG risks, seizing directs resources towards oversight and workflow redesign and transforming embeds these adjustments in organisational routines. Thus, GenAI is used not only to accelerate output but also to improve transparency, accountability and sustainability. DDC is expected to mediate only part of the GenAI–sustainability relationship because governance structures, stakeholder expectations and ethical orientation also shape sustainability outcomes. Therefore, this study proposes:

H3. Digital dynamic capability partially mediates the relationship between GenAI adoption and sustainability performance.

The relationship between GenAI adoption and sustainability is further conditioned by the extent to which firms institutionalise responsibility in AI governance. RAIO reflects commitment to fairness, transparency, accountability and human-centred design (Floridi and Cowls, 2022). When RAIO is strong, bias audits, environmental assessments and stakeholder consultation guide the reconfiguration enabled by DDC. When it is weak, the same capability may amplify bias, misinformation and ecological externalities. RAIO therefore does not eliminate the paradox; it directs adaptive reconfiguration towards responsible outcomes and strengthens DDC's mediating effect. Therefore, this study proposes:

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H4. Responsible AI orientation moderates the relationship between GenAI adoption and sustainability performance mediated by DDC such that the mediating effect is stronger when RAIO is high.

Figure 1 presents the proposed research model and the proposed hypotheses.

4. Study 1

4.1 Research objective

Study 1 estimates the short-term causal effects of GenAI adoption on market and sustainability signals. It tests whether adoption announcements generate abnormal returns and shifts in ESG sentiment, capturing investor expectations of efficiency and stakeholder concern about ethical or environmental risks. Cumulative abnormal returns represent forward-looking market expectations rather than realised operational gains (MacKinlay, 1997). ESG sentiment is a text-derived reputational signal rather than a direct measure of sustainability performance (Bapat *et al.*, 2022). A DiD-PSM reduces selection bias and provides the causal anchor for the longitudinal analysis in Study 2.

4.2 Data and sample

GenAI adoption events were manually verified and coded by organisational task and technology modality. From 612 announcements, duplicate and speculative references were removed, leaving 500 verified events; only the first announcement per firm was retained. Propensity-score matching on firm size, leverage, digital R&D intensity, pre-event ESG score and stock-return volatility produced 250 treated and 250 control firms. A 30-day pre- and post-event window around the announcement ($t = 0$) captures anticipatory and lagged responses. Outcomes are cumulative abnormal returns and ESG sentiment derived through natural language processing. The control firms were broadly comparable with treated firms on observable pre-treatment characteristics. Event identification and sample construction are detailed in the Online Appendix (Section C1); data sources, the first-event rule, event-window justification and sample composition are provided in Section A5 and Tables OA2–OA3.

4.3 Variable operationalisation

GenAI adoption was operationalised as a binary indicator requiring cross-verified deployment in an identifiable organisational function, excluding symbolic announcements. Among treated firms, 31.2% of events concerned marketing and text generation, 24.8% customer service, 18.4% software and code assistance, 14.0% knowledge management and 11.6% design or image generation. DDC was the first principal component of digital R&D ratio, digital patent count and IT renewal rate, explaining 62.8% of the variance. RAIO was a context-weighted

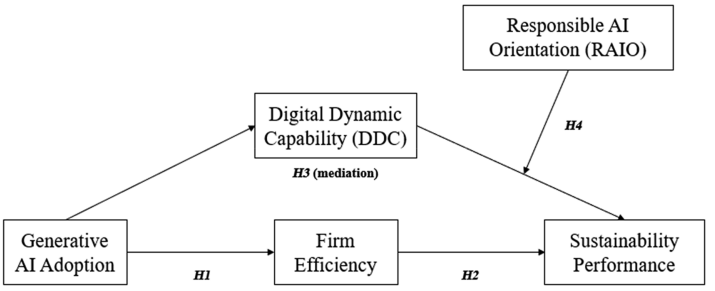


Figure 1. Proposed research model. **Source(s)**: Authors' own work

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index of action-oriented governance language validated through independent coding (Cohen's $\kappa = 0.81$; agreement = 87.2%). The DDC measure was also tested against an alternative archival proxy. Cumulative abnormal return captures investors' expectations rather than realised efficiency, while ESG sentiment captures stakeholders' evaluations rather than substantive sustainability performance. Measurement and validation details are provided in the Online Appendix (Sections A6, A8 and C2–C3; Tables OA4–OA5).

4.4 Model specification

The baseline DiD specification included firm and year fixed effects (FE) and explicitly controlled for firm size, leverage, firm age, pre-event ESG score, digital R&D intensity and stock-return volatility to account for firm-level heterogeneity. All models were estimated with firm-clustered robust standard errors to address within-firm serial dependence and heteroscedasticity. Full assumption tests, including residual diagnostics, heteroscedasticity tests, placebo analysis and leads-and-lags pre-trend tests, are reported in the Online Appendix (Sections C4–C5).

A difference-in-differences (DiD) model with firm and time FE estimates the causal impact of GenAI adoption on cumulative abnormal return and ESG sentiment (Imbens and Rubin, 2015).

$$Y_{it} = \alpha_i + \gamma_t + \beta_1 \mathrm{GenAI}_{it} + \beta_2 X_{it} + \varepsilon_{it}$$

GenAI_{it} equals 1 after adoption; the control vector X_{it} includes firm size, leverage, age and prior ESG. Propensity-score matching precedes DiD using logistic regression:

$$P(\mathrm{Adopt}_i = 1) = \Lambda(\delta_0 + \delta_1 \mathrm{Size}_i + \delta_2 \mathrm{Industry}_i + \delta_3 \mathrm{PreESG}_i + \delta_4 \mathrm{InnovationIndex}_i)$$

Nearest-neighbour matching (calliper = 0.01) achieves balance (ΔSize = 0.03148; ΔInnovation = 0.01736; ΔESG = 0.03697; $p > 0.10$), ensuring comparability (Abadie and Cattaneo, 2018).

4.5 Robustness and diagnostic checks

Two robustness checks are undertaken: a placebo test and a leads-and-lags pre-trend test with firm-clustered standard errors throughout; detailed coefficients are reported in the Online Appendix (Section A6).

4.6 Results

The estimation results, reported in Table 1, indicate distinct short-term effects of GenAI adoption across financial and sustainability dimensions, consistent in magnitude and direction with evidence from digital innovation event studies in comparable contexts (Mikalef *et al.*, 2022; Wang and Wang, 2023).

The cumulative abnormal returns coefficient for GenAI adoption ($\beta_1 = 0.02741$, $p = 0.00293$) indicates a 2.741% increase, consistent with investors' positive reassessment of firm value after the announcement. The ESG sentiment coefficient ($\beta_1 = -0.01946$, $p = 0.00748$) indicates heightened stakeholder concern, not deterioration in substantive sustainability performance. Together, the coefficients show GenAI's double-edged reputational effect: markets reward expected efficiency, while sustainability-oriented stakeholders signal ethical concern. DDC is associated with partial ESG recovery ($\beta = 0.01097$, $p = 0.01642$), and RAIO attenuates the ESG sentiment decline ($\beta = 0.01236$, $p = 0.01988$). High-RAIO firms experience approximately half the ESG loss of low-RAIO firms. These findings answer RQ1 by demonstrating simultaneous efficiency expectations and sustainability concerns in the short run.

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Table 1. Results of difference-in-differences estimation

	Cumulative abnormal return	ESG sentiment index
GenAI adoption (β_1)	0.02741 (SE = 0.00861, p = 0.00293)	−0.01946 (SE = 0.00685, p = 0.00748)
Firm size	0.00369 (SE = 0.00219, p = 0.08712)	0.00608 (SE = 0.00291, p = 0.04133)
Leverage	−0.01682 (SE = 0.00703, p = 0.02072)	−0.02362 (SE = 0.00871, p = 0.01063)
Pre-event ESG	0.00827 (SE = 0.00411, p = 0.03244)	0.03572 (SE = 0.01024, p = 0.00083)
Firm age	−0.00113 (SE = 0.00102, p = 0.28579)	0.00208 (SE = 0.00113, p = 0.21286)
Industry FE	Included	Included
Year FE	Included	Included
Observations	9,874	9,874
Adjusted R-squared	0.21941	0.26138
F-statistic	11.57418 (p < 0.00100)	13.24753 (p < 0.00100)

Note(s): This table reports the results of the difference-in-differences estimation with firm and year fixed effects. The dependent variables are market-based and sustainability-based outcomes measured around verified GenAI adoption events. Standard errors are robust and clustered at the firm level. Positive coefficients on GenAI adoption indicate a short-term, market-perceived efficiency signal reflected in cumulative abnormal returns, while negative coefficients for ESG sentiment highlight stakeholder concerns regarding sustainability implications. Controls include firm size, leverage, age and prior ESG score

Source(s): Authors' own work

An event-study plot displaying the time-specific β coefficients and 95% confidence intervals around the GenAI adoption event (six weeks before and after $t = 0$) is provided in the Online Appendix (Figure OA1). The pattern shows that pre-event coefficients are statistically indistinguishable from zero, supporting the parallel-trends assumption. It also shows that post-event cumulative abnormal returns responses are significantly positive and peak at approximately two weeks after adoption, consistent with a short-run surge in market optimism surrounding GenAI adoption.

The two studies use different samples and outcomes because they address distinct temporal questions. Study 1 analyses 500 adoption events within a 30-day pre- and post-event window, using cumulative abnormal returns and ESG sentiment. Study 2 analyses 1,274 firm-year observations from 512 firms between 2016 and 2024, using a composite Sustainability Index. DiD-PSM isolates the treatment effect around a discrete announcement but cannot model multi-year persistence, mediation, or moderation. System GMM models these longer-term processes but cannot identify a sharply dated announcement effect. The studies therefore form a complementary temporal-triangulation strategy.

5. Study 2

5.1 Objective

Study 2 examines the dynamic and persistent effects of GenAI adoption on firms' long-term sustainability. As explained above, Study 1 and Study 2 form a temporal-triangulation design in which each study's sample, time horizon and outcome measure are matched to a distinct explanatory task: whereas Study 1 captured short-term market and reputational reactions around discrete adoption events, Study 2 uses its longitudinal panel to evaluate whether these impacts endure or fade over time, focussing on how firms internalise GenAI-driven efficiencies through adaptive capabilities and responsible governance (Dwivedi *et al.*, 2023; Mikalef *et al.*, 2022). Using a two-step system GMM estimator (Arellano and Bover, 1995; Blundell and Bond, 1998), it models adjustment processes, learning effects and temporal dependence. System GMM corrects for endogeneity from lagged sustainability and bidirectional causality between performance and adoption. The estimator is suitable for short, unbalanced panels and efficiently handles heteroscedasticity (Roodman, 2009). An autoregressive coefficient of approximately 0.4–0.6 would confirm path dependence,

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indicating sustainability evolves cumulatively through organisational learning and responsible governance.

5.2 Data

The analysis uses an unbalanced panel of 1,274 firm-year observations from 512 firms in IT, telecommunications, e-commerce and fintech between 2016 and 2024, with data from Orbis and Refinitiv. GenAI adoption is identified from annual reports, ESG disclosures and press releases. Sustainability combines the Refinitiv ESG composite score and a text-derived stakeholder trust index. The first principal component explains 72.4% of the variance. The DDC index shows strong reliability ($\alpha = 0.823$, Kaiser–Meyer–Olkin [KMO] measure = 0.781), and RAIO demonstrates cross-year stability ($\rho = 0.618$, $p < 0.001$), supporting their interpretation as persistent organisational constructs.

5.3 Model specification

A two-step System GMM estimator (Arellano and Bover, 1995; Blundell and Bond, 1998) models the dynamic effect of GenAI adoption on sustainability while addressing endogeneity, reverse causality and firm heterogeneity. Lagged sustainability ($SUS_{i,t-1}$) captures persistence; GenAI adoption, DDC and the GenAI × RAIO interaction are the focal predictors. Lagged values of GenAI, DDC, R&D, digital capital and the interaction term serve as instruments, as detailed in Online Appendix Section A7. Instruments were collapsed to limit proliferation; the total count of 38 is below the number of firms ($N = 512$). Diagnostics support the specification (AR(1): $z = -2.614$, $p = 0.009$; AR(2): $z = -1.327$, $p = 0.185$; Hansen $J = 28.41$, $p = 0.328$; difference-in-Hansen: $p = 0.298$). Two-step standard errors are Windmeijer (2005)-corrected.

5.4 Results

Table 2 presents the estimation results of the two-step system GMM model. The findings reveal that sustainability performance exhibits moderate persistence, confirming its dynamic nature and gradual accumulation over time.

The autoregressive coefficient ($\alpha = 0.52438$) reveals strong state dependence: about half of a firm's current sustainability performance derives from the prior year, confirming sustainability as an accumulative, learning-based capability (George *et al.*, 2021). GenAI adoption intensity ($\beta_1 = 0.14627$, $p = 0.02014$) is associated with a positive but diminishing change in sustainability performance, reflecting an efficiency bias where early gains plateau amid ethical and environmental strain (Bengio *et al.*, 2024). DDC ($\beta_2 = 0.17392$, $p = 0.00168$) is associated with a significant mediating pathway. Given that the direct effect of GenAI adoption on sustainability remains statistically significant, the results indicate complementary partial mediation, whereby DDC enhances but does not fully account for the relationship. The GenAI by RAIO interaction ($\beta_3 = 0.11248$, $p = 0.01564$) confirms that RAIO mitigates ESG volatility through ethical oversight. The indirect effect (estimate = 0.03041, 95% confidence interval [0.01132, 0.04985]) validates partial mediation, underscoring that GenAI adoption is associated with higher sustainability performance only when coupled with digital adaptability and responsible governance. This analysis addresses RQ2 and RQ3 by showing that DDC partially mediates the GenAI-sustainability relationship and that RAIO strengthens the responsible conversion of GenAI-enabled efficiency into sustainability performance.

5.5 Robustness analyses

Three alternative operationalisations were tested to assess sensitivity to measurement choices: a conservative GenAI adoption intensity threshold (at least three deployment-specific references per firm-year), a non-textual DDC specification (digital R&D ratio and patent count only) and a narrow RAIO governance-only dictionary. In all three cases, as well as when firm-

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Table 2. Results of two-step system GMM estimation

Variable	Coefficient (β)	Standard error	z-Statistic	p-Value
$SUS_{i,t-1}$	0.52438	0.09146	5.73284	0.00000
$GenAI_{it}$	0.14627	0.06295	2.32316	0.02014
DDC_{it}	0.17392	0.05538	3.14187	0.00168
$GenAI_{it} \times RAIO_{it}$	0.11248	0.04651	2.41752	0.01564
Constant	0.03479	0.01967	1.76835	0.07698
Hansen J-test (p-value)	0.32781			
AR(2) test (p-value)	0.18452			
Difference-in-Hansen (p-value)	0.29793			
Observations	1,274			
Number of firms	512			

Note(s) (1) This table presents results from the two-step system generalised method of moments (GMM) estimator following Arellano–Bover and Blundell–Bond specifications
(2) The dependent variable is sustainability (SUS), defined as the principal component of the ESG composite and stakeholder trust index
(3) Lagged sustainability ($t - 1$) captures persistence ($\beta = 0.524$, $p < 0.001$)
(4) The GenAI × RAIO interaction indicates that responsible AI orientation strengthens the translation of digital efficiency into sustainable performance
(5) Instruments were collapsed using the xtabond2 "collapse" option in Stata 17 to mitigate instrument proliferation. The total instrument count is 38, below the number of firms ($N = 512$)
(6) Diagnostics support the specification: AR(1) is expected after first-differencing ($z = -2.614$, $p = 0.009$). AR(2) indicates no second-order autocorrelation ($z = -1.327$, $p = 0.185$). The Hansen J-test supports joint instrument validity ($J = 28.41$, $p = 0.328$). The difference-in-Hansen test supports the additional moment conditions ($p = 0.298$)
(7) Two-step standard errors are Windmeijer (2005)-corrected
Source(s): Authors' own work

years in the top decile of disclosure length were excluded, the core results remained positive, significant and consistent in magnitude. Full robustness analyses, including detailed coefficients and supporting discussion, are reported in the Online Appendix (Section A8 and Table OA6).

6. Discussion

6.1 General discussion

The two studies show that the sustainability trajectory of GenAI adoption develops through short-term efficiency expectations and longer-term governance adaptation. Study 1 identifies a positive market response alongside declining ESG sentiment, indicating investor optimism and stakeholder concern rather than realised efficiency or substantive sustainability deterioration. Study 2 shows that this imbalance can be partially stabilised as firms develop DDC and institutionalise RAIO. Sustainability persistence ($\alpha \approx 0.52$) indicates cumulative learning, but the underlying paradox is not eliminated. DDC helps firms realign innovation with responsibility, while RAIO anchors experimentation in fairness and accountability. Together, they reduce the risk that short-term efficiency gains become sustained legitimacy losses.

A detailed dynamic-capability-lens discussion of DDC and RAIO under GenAI is provided in the Online Appendix (Section A9).

6.2 Theoretical contributions

This study advances theory by integrating dynamic capability and responsible governance perspectives to explain the sustainability of GenAI adoption. Its primary theoretical contribution is not to show that the GenAI sustainability paradox is resolved but to explain how it may be mitigated and partially stabilised over time through the joint operation of DDC

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and RAIO. GenAI adoption is thus reframed not as a technological choice but as a capability-building process in which adaptability and accountability co-evolve. DDC enables firms to reconfigure digital resources amid algorithmic turbulence, while RAIO embeds ethical reflexivity that prevents efficiency-driven exploitation. Together, these mechanisms explain how firms manage a persistent tension between efficiency and responsibility rather than eliminate it.

A detailed elaboration of the contribution to dynamic capability theory and to responsible AI research is provided in the Online Appendix (Section A10). A further elaboration of the bright-dark duality as an evolving organisational tension, and a methodological note on the temporal-triangulation design, is provided in the Online Appendix (Section A11).

The moderation finding is particularly relevant to digital platforms and online service providers facing efficiency and legitimacy pressures. Prior Internet Research scholarship highlights the governance responsibilities of digital actors (Abedin, 2022; Birkstedt *et al.*, 2023). The present results show that transparency mechanisms, bias audits and human oversight function not only as ethical compliance but also as strategic moderators of whether digital efficiency produces durable legitimacy or reputational erosion.

6.3 Practical implications

A summary of practitioner-oriented guidance for managers, AI governance teams, regulators and investors linked to the study's core insight that GenAI sustainability depends on the co-evolution of dynamic capability (DDC) and responsible governance (RAIO) is provided in the Online Appendix (Table OA7).

From a managerial perspective, firms should build digital sensing, experimentation and knowledge recombination routines as the foundation of sustainable competitiveness. DDC should be viewed as a tangible capability manifested through iterative experimentation and resource orchestration. Building DDC entails investing in flexible digital infrastructures, fostering collaboration between data scientists and strategists and institutionalising learning from algorithmic feedback loops. As Mikalef *et al.* (2022) argued, the ability to reconfigure technological assets amid turbulence distinguishes sustainable advantage from inertia. As GenAI delivers short-term efficiency but can erode long-term legitimacy, embedding feedback cycles in which algorithmic outputs are reviewed and retrained by human experts helps transform efficiency into adaptability and align digital agility with ethical integrity.

For AI governance teams, RAIO emerges as an operational necessity rather than symbolic compliance. Responsible AI requires systematic bias audits, energy transparency and algorithmic documentation. Bias audits ensure representational fairness (Floridi and Cowls, 2022), while energy transparency links GenAI use to ESG scrutiny (Bengio *et al.*, 2024). Embedding RAIO metrics, including fairness, explainability and environmental efficiency, into corporate dashboards integrates responsibility into strategic control systems, providing early warnings for ethical deviations. At the regulatory level, the results support adaptive governance that balances innovation with accountability. Regulators should establish a "Responsible GenAI Certification," akin to International Organization for Standardization (ISO) 14001 or General Data Protection Regulation (GDPR), covering fairness, data provenance and transparency. Consistent with Dwivedi *et al.* (2023), multi-stakeholder collaboration can harmonise governance and prevent fragmentation. Proportional oversight, stringent for high-risk sectors and lighter for low-risk uses, preserves both agility and legitimacy. For investors, the positive link between DDC, RAIO and sustainability suggests that firms combining agility with ethical oversight are better positioned to sustain positive ESG trajectories and resilient long-term returns.

6.4 Limitations and future research

The findings should be interpreted within the study's empirical scope. Its focus on digitally intensive sectors limits generalisability to traditional industries. Firm-level adoption also

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masks differences across tasks and technologies, including model opacity, energy intensity, customer-facing deployment and human oversight. Moreover, the documented efficiency-responsibility tension is specific to generative systems' production of novel synthetic content and associated risks of hallucination, authenticity and transparency; it should not be generalised uncritically to predictive AI. Future research should compare task domains and industries and combine archival measures with surveys, internal audits or experiments. Additional directions concerning RAIO measurement, energy life-cycle assessment and comparative regulation are discussed in Online Appendix Section A12.

Further discussion of methodological limitations and future research directions, including RAIO measurement, energy life-cycle assessment and comparative regulatory analysis, is provided in the Online Appendix (Section A12).

Supplementary material

The supplementary material for this article can be found online.

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